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Industry Analysis

In the 20th century the invention of the computer was a massive step forward in technology. This invention was followed by the creation and rapid expansion of the internet, as well as the proliferation of hardware devices such as smartphones and tablets all interconnected by the internet. The result was not just a huge change in the lives of everyday people, but a massive expansion in the Computer Software Industry. According to statista.com, “In 2015, 86.8 percent of all households in the United States had a computer at home.” (U.S. households with PC/computer n.d.). All of these computers have software installed on them to perform a huge variety of tasks related to business, entertainment, and social media. In the past few decades the world has seen many technology giants emerge and all these tech giants need talented and skilled workers to make their software a reality. According to the United States Department of Labor Bureau of Labor Statistics, there were 1,256,200 Software Developer jobs in the year 2016, and furthermore, the job outlook for the future is quite great: “Employment of software developers is projected to grow 24 percent from 2016 to 2026, much faster than the average for all occupations.” (Summary. 2018, April 13.)

So, what does a Software Engineer do? The Department of Labor states: “Software developers are the creative minds behind computer programs. Some develop the applications that allow people to do specific tasks on a computer or another device. Others develop the underlying systems that run the devices or that control networks.” (Summary. 2018, April 13.) This is a

suitable statement for what Software Engineers do in their day to day from a high level. More specifically, the types of applications and networks that these Software Engineers are working on have evolved over time. Today, Software Engineers are writing code for web applications, mobile applications, and desktop applications. Depending on the type of application, the code will be in a different programming language. For example, Android phone applications are written in the Java programming language or Kotlin, whereas IOS applications are written in Swift or Objective-C. Web development is done traditionally in some combination of HTML, CSS, JavaScript frameworks, and Server-side frameworks such as PHP or C# in the .Net Framework. This demonstrates the variety in types of work done by Software Engineers. There are many different avenues that one can choose when they decide to become a Software Engineer. That being said, there are still many commonalities between the different development types. The Object-Oriented style of computer programming and the Model-View-Controller design pattern serve as guidelines across multiple paradigms.

The big companies in the computer software industry include Microsoft, Oracle, SAP, IBM, Google, Facebook, and Apple, and many others. Microsoft is a computer software products and services company whose main product is the Windows operating system, a ground-breaking product that introduced the graphical user interface to computers.

On April 4, 1975, Bill Gates and Paul Allen founded Microsoft to create software for an electronics company called MITS. MITS was the manufacturer of a computer called the Altair 8800, and Microsoft originally developed software for this computer. (Microsoft founded. 2015). Microsoft released MS-DOS in 1981, it's first operating system, and then its famous Windows operating system in 1985, which was acclaimed for its graphical user interface. In the 90s,

Microsoft's Windows 95 release was extremely successful, and since then Microsoft has also made forays into web browsing and video games. (Microsoft founded. 2015).

Bill Gates is the Former CEO and co-founder of Microsoft. Gates developed most of Microsoft's most influential software, the GUI-based Windows operating system. Gates co-founded Microsoft with Paul Allen, his childhood friend after dropping out of Harvard University in 1975. (Microsoft founded. 2015). Paul Allen was a co-founder of Microsoft with Bill Gates. Allen was a childhood friend of Gates who quit his job to join Gates in starting a software company. Allen stayed on with Microsoft only until 1983 when he had health difficulties with Hodgkin's lymphoma. (Microsoft founded. 2015). Steve Ballmer was a CEO of Microsoft who went to Harvard University where he met Bill Gates during the year Gates founded Microsoft. He was the 30th employee to join Microsoft. (Mazarkis. 2015). Ballmer cites his best early success as being the establishment of the college recruiting system and discusses the difficulty in being the boss of Gates when Gates served as Chief Software Architect before leaving in 2008. (Mazarakis, personal communication, 2017).

The legacy of these three individuals, and of many others who have worked for Microsoft, is one of the most successful and well known software companies in the world. Microsoft was one of the pioneers of the graphics-based computer operating systems. The Windows operating system and the Mac OS operating system were essentially the two biggest competitors on the market. What set the Windows operating system aside from what was available at the time was its Graphical User Interface, or GUI. Microsoft's first operating system, MS-DOS, did not have a GUI. It was all text based. In order to get the computer to run programs, the user had to type text based commands in. In order to navigate the file system, the user had to do the same. What Windows offered graphics to represent files. According to Ward, "Windows!

This was the first version of a Microsoft OS that you didn't have to type in commands to use. Instead, you could point and click in a box — a window — with a mouse.” (Ward, n.d.)

The products that Microsoft offers include the Windows operating systems, the Microsoft Office software suite, the Internet Explorer and Edge web browsers, the Xbox video game consoles, the Surface touchscreen devices, the Microsoft .NET Framework, and much more that can be found on their website. (Microsoft Products, n.d.). The Xbox line of video game consoles has been very successful. The .NET Framework is Microsoft’s main suite of Software Development tools. According to Geeks for Geeks, “.NET is a software framework which is designed and developed by Microsoft. The first version of .Net framework was 1.0 which came in the year 2002. In easy words, it is a virtual machine for compiling and executing programs written in different languages like C#, VB.Net etc.” (Introduction to .NET Framework. 2018).

Working at Microsoft is generally considered to be a positive experience. Microsoft has a 4.1 out of 5 rating on Glassdoor.com on over 18840 reviews which illustrates Microsoft is a positive company for employee experience. (Working at Microsoft. n.d.). The percentage of employees who approve of the CEO is exceptionally high at 96%, 84% of the reviewers recommend working at Microsoft to their friends, and the CEO was rated as one of the top CEO’s of 2018 by employees.

The newest developments in Software Engineering include Artificial Intelligence, Mobile/Web development, and Cybersecurity. (Williams, J. 2018). All these areas are having substantial and exciting growth. Artificial Intelligence, for example, has applications in many different industries such as finance, energy, eCommerce, and automation. In Web development, there are newer technologies that provide better solutions for mobile adaptive user interfaces with data storage in the cloud, so users can take advantage of applications on the go without

worrying about losing their data. Software developers can expect to be a part of a lot of exciting work in the future. The growth in these areas ensure that Software Engineers will be in high demand for years to come.

The jobs that are particularly interesting in the software industry are Software Engineer, Software Architect, and CTO, or Chief Technology Officer. As discussed earlier, a Software Engineer's main roles involve writing code and creating the applications that the customers use. They are on the front lines of development. A Software Architect, however, is a more high-level position and more of a management/business position. According to Korkishko, "The most important responsibility is the *complete technical support* of the project from the moment of inception, through product release, to development of enhancements." (2018). He also lists more responsibilities such as identifying business and stakeholder requirements, designing an entire system based on requirements, holding code reviews, choosing the technologies to use, and more. Software Architects need to have more than just programming skills, they need to be effective leaders and communicators and they need to have strong management skills. They are the interface between the programmers and the business people, and they are responsible for figuring out which technologies and application designs will best suit the needs of the business and the client. This means the Software Architect has much more responsibility and level of control over the application design cycle.

Going even further, there is the Chief Technology Officer position, or CTO. According to Frankenfield, "A chief technology officer (CTO) is the executive in charge of an organization's technological needs as well as its research and development (R&D). Also known as a chief technical officer, this individual examines the short- and long-term needs of an organization and utilizes capital to make investments designed to help the organization reach its objectives."

(2018). The position of CTO is one that is highly valued, and a position that an ambitious young Computer Science major might envy having in the future. The position comes with a level of responsibility and impact on the business that lower level positions do not have.

In order to get a job in the software industry as a Software Engineer it is generally recommended that one has at least a bachelor's degree in computer science. For the higher-level positions of Software Architect and Chief Technology Officer, higher levels of education are justifiably considered more important. That being said, there has been a movement recently toward considering talent that does not have college degrees and evaluating them solely on the basis of their programming skills. Bill Gates, after all, was a college dropout. However, the four-year degree is still highly valued, and any potential candidate for a Software Engineering position will highly benefit from having the degree, the knowledge, and experience that comes from obtaining one.

Outside of the college degree and college experience, there are many other avenues that one can take that can lead to successfully landing a job. These include certifications and online code training. An example of online certification would be the company Udacity. Udacity offers what they call Nanodegrees, which contain courses that teach technology concepts and certify that the enrollee is learning the concepts well enough to be awarded a certification at the end. Additionally, Udacity offers career support and networking for those that do well in the program, even offering to help find the student a job. For example, the Android and IOS nanodegrees teach mobile application development while having the learner develop portfolio worthy applications. The Android nanodegree is even written by Google engineers, which is especially beneficial since Google backs Android development. (Free Online Classes & Nanodegrees, n.d.)

In addition to certifications, there are online sites that teach computer programming, Web development, and Mobile development. A few examples of these sites would be Udemy, Tutorials Point, and W3Schools. There are even a lot of free tutorials on YouTube that one can view and learn from. While they may not offer official certifications that a company could verify during a job search, these sites are excellent resources to use as references for a Software Engineer. Most Web Developers, for example, won't have the exact syntax of a JavaScript split function for splitting a string memorized. The exact syntax and usage of such a function is available on W3Schools' website. This makes W3Schools' website an excellent reference to have on hand during development, or for preparation before a job interview. (JavaScript String split() Method. (n.d.)).

In addition to the technical preparation one would need to do in order to become employed as a Software Engineer, there is also the importance of building an impressive resume. A lot of employers look for candidates that have years of experience, even for entry level positions. Thus, it is important for young and aspiring Software Engineers to develop their resumes before they attain job experience. ComputerScience.org has several recommendations for undergraduates planning on becoming a Software Engineer. These include joining clubs on campus, "seek[ing] out internships whether paid or unpaid for infield experience", tutoring fellow students, and "offering free coding services to friends and businesses [to] build up your resume." (Software Engineer Careers. n.d.). Not only will coding a lot of applications and different projects lead to an increase in skill level, but they have an added effect of being things that can be added to a resume. For example, having applications released to the Google Play Store or to the iPhone App Store shows a potential employer that one has experience in taking an application through development all the way to release.

Finally, there are online resources for finding employment. LinkedIn, for example, is widely known as being used for online networking and job searching. There are also other websites such as Dice.com, Stackoverflow.com, and Glassdoor.com. One of the best ways for a prospective Software Engineer to gain exposure, market themselves, market their portfolio, and land a job is to create accounts on all of these sites and upload a resume. Begin responding to as many listings as possible that are within the skill and experience level and make that push for an interview. From there, the sky is certainly the limit for what one can achieve.

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